

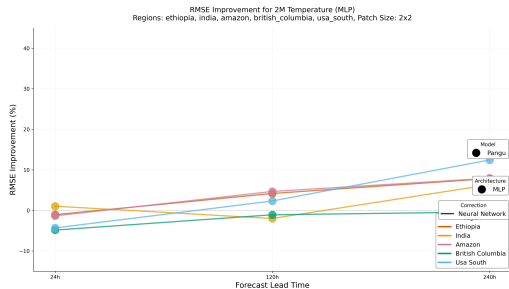
Scratch Work: Tailoring Machine Learning Weather Predictions for Local Impacts

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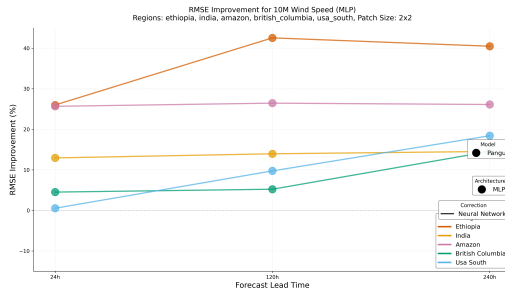
What's new?

- Rewrote databuild to be much faster and allow for using more regions. Early results for Ethiopia
- Added month embeddings to unet and mlp architectures.
- Capable to train on multiple lead times at once using learned lead time embedding.
- Implemented Bayesian hyperparameter optimization and optimized parameters for 2m temperatures 1 week lead time in India.

Do Better with Temperature: 2x2

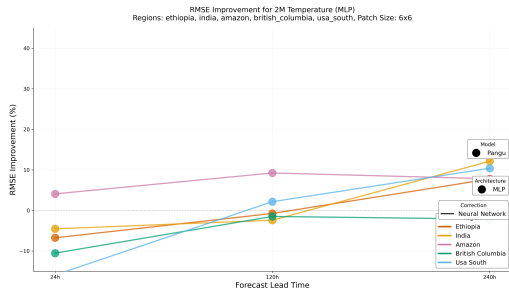


(a) 2m Temperature: Pangu

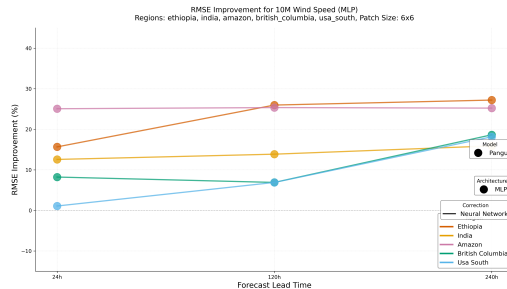


(b) Wind Speed: Pangu

6x6 Region Size

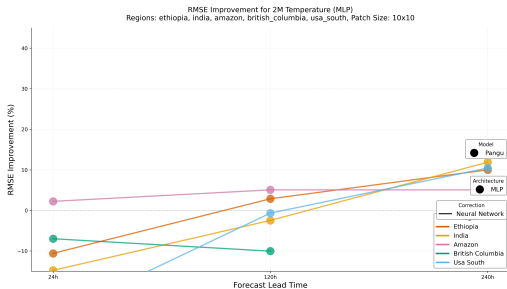


(a) 2m Temperature: Pangu

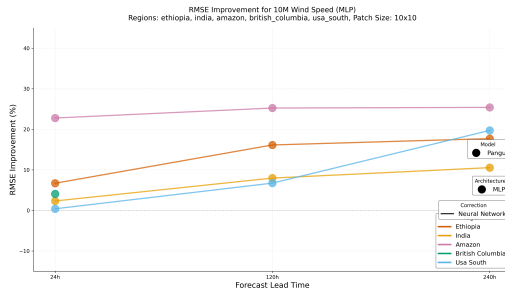


(b) Wind Speed: Pangu

10x10 Region Size



(a) 2m Temperature: Pangu



(b) Wind Speed: Pangu

Next Steps

- Finish script that downloads graph/genecast from GEE
- Get AIFS and IMD from Pedram's group
- More thorough hyperparameter optimization
- A few things I can do to speed up training and databuild